
Entraînement au calcul algébrique.

Les résultats généraux obtenus dans ces trois premières questions sont à connaître.

Les divers paramètres qui apparaissent $(a, b, c, x, y, \dots)$ sont des réels.

La calculatrice n'est pas autorisée.

Autour des puissances de -1

Question 1. Trouver une expression plus simple valable pour tout $n \in \mathbb{N}$:

a) $\frac{1}{(-1)^n}$ b) $(-1)^{n+2}$ c) $(-1)^{2n}$ d) $(-1)^{2n+1}$

Développements - factorisations

Question 2. 1°) Rappeler les identités remarquables pour $(a+b)^2$, $(a-b)^2$, $(a+b)(a-b)$.

2°) Développer et simplifier $(a+b+c)^2$.

Question 3. 1°) Développer $(a-b)(a^2+ab+b^2)$.

2°) En déduire (directement, sans faire "d'analogie") une factorisation de a^3+b^3 .

3°) Application : Factoriser $27x^3+8$.

Règles des puissances de -1

Question 1:

$$\begin{aligned} \text{a) } \frac{1}{(-1)^n} &= (-1)^n \quad \times \\ \text{b) } (-1)^{m+2} &= (-1)^m \times (-1)^2 \\ &= (-1)^m \quad \checkmark \\ \text{c) } (-1)^{2n} &= ((-1)^2)^n \\ &= 1^n = 1 \quad \times \\ \text{d) } (-1)^{2n+1} &= ((-1)^2)^n \times (-1)^1 \\ &= 1^n \times (-1) \\ &= -1^n = -1 \quad \times \end{aligned}$$

$\underbrace{\hspace{10em}}_{1^n = 1}$

Développements - factorisations

Question 2:

$$1) (a+b)^2 = a^2 + 2ab + b^2 \quad \checkmark; (a-b)^2 = a^2 - 2ab + b^2 \quad \checkmark; (a-b)(a+b) = a^2 - b^2 \quad \checkmark$$

$$2) (a+b+c)^2 = (a+b+c)(a+b+c)$$

$$= \cancel{a^2} + \cancel{ab} + \cancel{ac} + \cancel{ab} + \cancel{b^2} + \cancel{bc} + \cancel{ac} + \cancel{bc} + \cancel{c^2}$$

$$= a^2 + 2ab + b^2 + 2bc + c^2 + 2ac$$

$$= a^2 + b^2 + c^2 + 2(ab + ac + bc) \quad \checkmark$$

Question 3:

$$1) (a-b)(a^2 + ab + b^2) = a^3 + \cancel{a^2b} + \cancel{ab^2} - \cancel{a^2b} - \cancel{ab^2} - b^3 \\ = a^3 - b^3 \quad \checkmark$$

$$2) \text{ On aurait } a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad \times$$

$$3) 27x^3 + 8 \text{ On a ici : } b^3 = 8 \text{ donc } b=2 \text{ et } a^3 = 27x^3 \text{ donc } a=3x$$

$$\text{N'ai } 27x^3 + 8 = (3x+2)((3x)^2 - 3x \times 2 + 2^2) \\ = (3x+2)(9x^2 - 6x + 4) \quad \times$$

Entraînement au calcul algébrique : feuille 1.

Les divers paramètres qui apparaissent ($a, b, c, x, y, \dots$) sont des réels.

La calculatrice n'est pas autorisée.

Factorisations

Question 4. Les expressions suivantes peuvent se factoriser sans jamais développer, soyez bien attentif pour repérer les facteurs communs.

1°) $A = x^4 - x^2$

2°) $B = x^2 - 2x + 1 - (x - 1)(2x + 3)$

3°) $C = (3x + 2)^2 - (x - 1)^2$

4°) $D = x^3 + x^2 + x + 1$

5°) $E = 9x^2 - 49 + (3x + 7)(2x + 3)$

Quotients

On ne se préoccupera pas de l'existence lorsqu'il y a des variables indéterminées.

Question 5. Réduire (de la manière la plus efficace) :

$$\frac{7}{8} + \frac{5}{12} - \frac{4}{3} \qquad \frac{1}{(n+1)^2} + \frac{1}{n+1} - \frac{1}{n}$$

Question 6. Simplifier, sans se préoccuper de l'existence :

$$x = \frac{\frac{a}{b}}{\frac{c}{d}}, \quad y = \frac{\frac{a}{b}}{\frac{c}{d}}, \quad z = \frac{\frac{a}{b}}{\frac{c}{d}}, \quad t = \frac{\frac{a}{b}}{\frac{c}{d}}$$

$$A = \frac{\frac{a^2}{3b}}{\frac{6b}{ac}}, \quad B = \frac{\frac{3}{10} \times \frac{15}{9} \times \frac{2}{5}}{\frac{15}{2}}, \quad C = \frac{6(3 - \frac{1}{2})(4 + \frac{1}{3})}{12(5 + \frac{1}{4})(7 - \frac{1}{3})}$$

Question 7. Simplifier, sans se préoccuper de l'existence :

$$A = \frac{1}{x-1} - \frac{1}{x+1} + \frac{2x}{1-x^2} \qquad B = \frac{x-a}{ax} + \frac{a-b}{ab} + \frac{b-x}{xb}$$

Factorisation

question 4:

$$\begin{aligned} 1) A &= x^4 - x^2 \\ &= x^2(x^2 - 1) \\ &= x^2(x+1)(x-1) \checkmark \end{aligned} \quad \begin{aligned} 2) B &= x^2 - 2x + 1 - (x-1)(2x+3) \\ &= (x-1)^2 - (x-1)(2x+3) \\ &= (x-1)(x-1-2x-3) \\ &= (x-1)(-x-4) \checkmark \end{aligned} \quad \begin{aligned} 3) C &= (3x+2)^2 - (x-1)^2 \\ &= (3x+2-x+1)(3x+2+x-1) \\ &= (2x+3)(4x+1) \checkmark \end{aligned}$$

$$\begin{aligned} 4) D &= x^3 + x^2 + x + 1 \\ &= x^2(x+1) + x+1 \\ &= (x+1)(x^2+1) \checkmark \end{aligned} \quad \begin{aligned} 5) E &= 9x^2 - 49 + (3x+7)(2x+3) \\ &= (3x-7)(3x+7) + (3x+7)(2x+3) \\ &= (3x+7)(3x-7+2x+3) \\ &= (3x+7)(5x-4) \checkmark \end{aligned}$$

Quotients:

Question 5:

$$\begin{aligned} \frac{7}{8} + \frac{5}{12} - \frac{4}{3} &= \frac{21}{24} + \frac{10}{24} - \frac{32}{24} \\ &= -\frac{1}{24} \checkmark \end{aligned}$$

$$\begin{aligned} \frac{1}{(m+1)^2} + \frac{1}{m+1} - \frac{1}{m} &= \frac{1}{(m+1)^2} + \frac{m+1}{(m+1)^2} - \frac{1}{m} \\ &= \frac{m+2}{(m+1)^2} - \frac{1}{m} \\ &= \frac{m^2+2m-(m+1)^2}{m(m+1)^2} \\ &= \frac{\cancel{m^2}+\cancel{2m}-\cancel{m^2}-\cancel{2m}-1}{m(m+1)^2} \\ &= -\frac{1}{m(m+1)^2} \end{aligned}$$

Question 6:

$$x = \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

$$y = \frac{a}{\frac{b}{\frac{c}{d}}} = \frac{a}{b \times \frac{d}{c}} = \frac{a}{\frac{bd}{c}} = a \times \frac{c}{bd} = \frac{ac}{bd} \checkmark$$

$$J = \frac{\frac{a}{\frac{b}{c}}}{d} = \frac{a \times \frac{c}{b}}{d} = \frac{\frac{ac}{b}}{d} = \frac{ac}{b} \times \frac{1}{d} = \frac{ac}{bd} \quad \checkmark$$

$$L = \frac{\frac{\frac{a}{b}}{c}}{d} = \frac{\frac{a}{b} \times \frac{1}{c}}{d} = \frac{\frac{a}{bc}}{d} = \frac{a}{bc} \times \frac{1}{d} = \frac{a}{bcd} \quad \checkmark$$

$$A = \frac{\frac{a^2}{3b}}{\frac{ac}{6b}} = \frac{a^2}{3b} \times \frac{6b}{ac} = \frac{a^2 \cancel{6b}}{3b \cancel{a} c} = \frac{6a}{3c} = \frac{2a}{c} \quad \checkmark$$

$$B = \frac{\frac{3}{10} \times \frac{15}{2}}{\frac{9}{15} \times \frac{5}{2}} = \frac{\frac{3}{10} \times \frac{30}{2}}{\frac{9}{15} \times \frac{5}{2}} = \frac{\frac{\cancel{3} \times \cancel{3} \times 10}{\cancel{10} \times \cancel{5} \times 2}}{\frac{\cancel{9} \times \cancel{5} \times 2}{\cancel{3} \times \cancel{5} \times 2}} = \frac{1}{\frac{3}{2}} = 1 \times \frac{2}{3} = \frac{2}{3} \quad \checkmark$$

$$C = \frac{6(3 - \frac{1}{2})(4 + \frac{1}{3})}{12(5 + \frac{1}{4})(7 - \frac{1}{3})} = \frac{6(\frac{6}{2} - \frac{1}{2})(\frac{12}{3} + \frac{1}{3})}{12(\frac{20}{4} + \frac{1}{4})(\frac{21}{3} - \frac{1}{3})} = \frac{1}{2} \times \frac{\frac{5}{2} \times \frac{13}{3}}{\frac{21}{4} \times \frac{20}{3}} = \frac{1}{2} \times \frac{\frac{\cancel{5} \times 13}{\cancel{2}}}{\frac{21 \times 4 \times \cancel{5}}{2}} = \frac{13}{21 \times 4} = \frac{13}{84} \quad \checkmark$$

Question 7:

$$A = \frac{1}{x-2} - \frac{1}{x+1} + \frac{2x}{1-x^2} = \frac{\cancel{x+1} - \cancel{x} + 1}{(x-1)(x+1)} + \frac{2x}{1-x^2} = \frac{2}{(x-1)(x+1)} + \frac{2x}{(1-x)(1+x)} = \frac{2}{(x-1)(x+1)} + \frac{-2x}{-(1-x)(x+1)}$$

$$= \frac{2}{(x-1)(x+1)} + \frac{-2x}{(x-1)(x+1)}$$

$$= \frac{2-2x}{(x-1)(x+1)}$$

$$= \frac{2(1-x)}{(x-1)(x+1)}$$

$$B = \frac{x-a}{ax} + \frac{a-b}{ab} + \frac{b-x}{xb} = \frac{b(x-a) + x(a-b) + a(b-x)}{a \times b} = \frac{2(x-1)}{(x-1)(-1-x)}$$

$$= \frac{2}{-1-x} \quad \checkmark$$

$$= \frac{\cancel{bx} - ab + \cancel{ax} - \cancel{xb} + ab - \cancel{ax}}{a \times b}$$

$$= 0 \quad \checkmark$$

Puissances

Question 8. Mettre sous forme irréductible : $\frac{14^2 \times 9^2}{3^5 \times 7}$

Question 9. Mettre les nombres suivants sous la forme $2^a 3^b$ où a et b sont entiers.

$$x = \frac{2^3 3^2}{6^{-2} 3^4 2^8}, \quad y = 2^{100} + 2^{101}, \quad z = 2^{101} - 2^{100}, \quad t = 3^{15} + 3^{15}, \quad u = \frac{(3^2(-2)^4)^8}{((-3)^5 2^3)^{-2}}$$

Question 10. Simplifier, sans se préoccuper de l'existence :

$$A = 3x^2 y^3 - y(xy)^2 \quad B = \frac{4x^2 y^3 - (xy)^2 y}{x^2 y^2 \times (-x)^3} \quad C = \frac{(-a)^7 \times (-b^2 c^3)^3}{-b^3 c \times (-a)^5}$$

Racines

Question 11. Simplifier, de sorte qu'aucune racine n'apparaisse au dénominateur :

$$A = \frac{4}{3 - \sqrt{5}}, \quad B = \frac{1}{\sqrt{\sqrt{2}}} \sqrt{\frac{1 + \sqrt{2}}{2}}, \quad C = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} - \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}.$$

Question 12. Simplifier, pour $x \in]-1, 1[$, $A = \frac{\frac{1}{1+x} + \frac{1-x}{(1+x)^2}}{\sqrt{\frac{1-x}{1+x} \left(1 + \frac{1-x}{1+x}\right)}}$.

Puissances

Question 8:
$$\frac{14^2 \times 9^2}{3^5 \times 7} = \frac{(7 \times 2)^2 \times 3^4}{3^5 \times 7} = \frac{7^2 \times 2^2 \times 3^4}{3^5 \times 7} = \frac{7 \times 2^2}{3} = \frac{28}{3} \checkmark$$

Question 9:

$$\begin{aligned} x &= \frac{2^3 3^2}{6^{-2} 3^4 2^8} \\ &= \frac{2^3 3^2}{2^{-2} 3^{-2} 3^4 2^8} \\ &= \frac{2^3 3^2}{2^6 3^2} \\ &= 2^{-3} \checkmark \end{aligned}$$

$$\begin{aligned} y &= 2^{100} + 2^{101} \\ &= 2^{100} + 2^{100} \times 2 \\ &= 2^{100} (1 + 2) \\ &= 3 \cdot 2^{100} \checkmark \end{aligned}$$

$$\begin{aligned} z &= 2^{101} - 2^{100} \\ &= 2 \times 2^{100} - 2^{100} \\ &= 2^{100} (2 - 1) \\ &= 2^{100} \checkmark \end{aligned}$$

$$\begin{aligned} t &= 3^{15} + 3^{15} \\ &= 2 \cdot 3^{15} \checkmark \end{aligned}$$

$$\begin{aligned} u &= \frac{(3^2 (-2)^4)^8}{((-3)^5 2^3)^{-2}} \\ &= \frac{(3^2)^8 \cdot ((-2)^4)^8}{((-3)^5)^{-2} \cdot (2^3)^{-2}} \\ &= \frac{3^{16} \cdot (-2)^{32}}{-3^{-10} \cdot 2^{-6}} \\ &= \frac{3^{16} \cdot 2^{36} \cdot (-1)^{32}}{3^{-10} \cdot 2^{-6} \cdot (-1)^{-10}} \\ &= 3^{26} \cdot 2^{38} \cdot (-1)^{42} \\ &= 3^{26} \cdot 2^{38} \checkmark \end{aligned}$$

Question 10:

$$\begin{aligned} A &= 3x^2 y^3 - y(xy)^2 \\ &= 3x^2 y^3 - y \times x \times y \times x \times y \\ &= 3x^2 y^3 - x^2 y^3 \\ &= x^2 y^3 (3 - 1) \\ &= 2x^2 y^3 \checkmark \end{aligned}$$

$$\begin{aligned} B &= \frac{4x^2 y^3 - (xy)^2 y}{x^2 y^2 \times (-x)^3} \\ &= \frac{4x^2 y^3 (4 - 1)}{x^2 y^2 \times (-x)^3} \\ &= -\frac{3y}{x^3} \checkmark \end{aligned}$$

$$\begin{aligned} C &= \frac{(-a)^7 \times (-b^2 c^3)^3}{-b^3 c \times (-a)^5} \\ &= \frac{(-a)^7 \times (b^2)^3 \times (c^3)^3}{-b^3 c \times (-a)^5} \\ &= -\frac{(-a)^2 \times (-b)^6 \times c^9}{b^3 \times c} \\ &= \frac{a^2 \times (-b)^6 \times c^8}{b^3} \\ &= \frac{a^2 \times b^6 \times c^8}{b^3} = a^2 b^3 c^8 \checkmark \end{aligned}$$

Racines

Question 11:

$$\begin{aligned} A &= \frac{4}{3-\sqrt{5}} \\ &= \frac{4(3+\sqrt{5})}{(3-\sqrt{5})(3+\sqrt{5})} \\ &= \frac{4(3+\sqrt{5})}{9-5} \\ &= \frac{4(3+\sqrt{5})}{4} \\ &= 3+\sqrt{5} \quad \checkmark \end{aligned}$$

$$\begin{aligned} B &= \frac{1}{\sqrt{2}} \cdot \sqrt{\frac{1+\sqrt{2}}{2}} \\ &= 2^{-\frac{1}{4}} \cdot \sqrt{\frac{1+\sqrt{2}}{2}} \\ &= 2^{-\frac{1}{4}} \cdot \frac{\sqrt{1+\sqrt{2}}}{\sqrt{2}} \\ &= 2^{-\frac{3}{4}} \times \sqrt{1+\sqrt{2}} \\ &= \frac{\sqrt{\sqrt{2}+2}}{2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} C &= \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} - \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} \\ &= \frac{(\sqrt{3}+\sqrt{2})^2 - (\sqrt{3}-\sqrt{2})^2}{(\sqrt{3}-\sqrt{2})(\sqrt{3}+\sqrt{2})} \\ &= \frac{3+2\sqrt{3}\sqrt{2}+2 - (3-2\sqrt{3}\sqrt{2}+2)}{3-2} \\ &= \frac{2+2\sqrt{3}\sqrt{2}+2 - 3+2\sqrt{3}\sqrt{2}-2}{3-2} \\ &= 2(2\sqrt{3}\sqrt{2}) \\ &= 4\sqrt{3}\sqrt{2} \\ &= 4\sqrt{6} \quad \checkmark \end{aligned}$$

Question 12: $\forall x \in]-1, 1[$

$$\begin{aligned} A &= \frac{\frac{1}{1+x} + \frac{1-x}{(1+x)^2}}{\sqrt{\frac{1-x}{1+x}} \left(1 + \frac{1-x}{1+x}\right)} \\ &= \frac{\frac{x}{(1+x)^2}}{\sqrt{\frac{1-x}{1+x}} \times \frac{2}{1+x}} \\ &= \frac{1}{(1+x)^2} \times \frac{(1+x)\sqrt{1+x}}{\sqrt{1-x}} \\ &= \frac{\sqrt{1+x}}{(1+x)\sqrt{1-x}} \\ &= \frac{\sqrt{(1+x)(1-x)}}{(1+x)(1-x)} \\ &= \frac{\sqrt{1-x^2}}{1-x^2} \\ &= \frac{1}{\sqrt{1-x^2}} \quad \checkmark \end{aligned}$$

Entraînement au calcul algébrique : feuille 2.

Quotients

Question 13. Simplifier :

$$A = \frac{\frac{1}{1 + \frac{1}{1 + \frac{1}{2}}}}{1 + \frac{1}{1 - \frac{1}{2}}} \quad (\text{on pourra faire des calculs intermédiaires}), \quad B = \frac{2 + \frac{2+a}{2-a}}{2 - \frac{2+a}{2-a}}$$

Question 14. Simplifier :

$$A = \frac{\left(\frac{1}{a} + \frac{1}{b} - \frac{x}{ab}\right)(x+a+b)}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{2}{ab} - \frac{x^2}{a^2b^2}} \quad B = \frac{2x+3}{2(2x-3)} + \frac{12x}{9-4x^2} + \frac{3-2x}{4x+6}$$

Puissances

Question 15. Écrire comme produit de puissances de nombres premiers : 7840.

Question 16. Simplifier, sans se préoccuper de l'existence :

$$A = \frac{-3\left(\frac{2}{3}\right)^2 + 8\left(\frac{7}{2}\right)^2}{5\left(\frac{2}{5}\right)^2 - 6\left(\frac{4}{3}\right)^2}$$

Question 17. Parmi les nombres suivants, lesquels peuvent s'écrire sous la forme a^n avec a et n entiers ($n \neq 1$) ? Le cas échéant, déterminer a et n , avec n le plus grand possible.

$$12^3 \times 3^3 \quad 125^2 \times 3^6 \quad 3^3 \times 5^6 \quad 7^2 \times 2^3$$

Développements

Question 18. 1°) Montrer que, pour a, b, c réels : $a^2(c-b) + b^2(a-c) + c^2(b-a) = (a-b)(b-c)(c-a)$.

2°) Montrer que, pour a, b, c réels : $(a+b+c)^2 + (-a+b+c)^2 + (a-b+c)^2 + (a+b-c)^2 = 4(a^2+b^2+c^2)$

Question 13:

$$A = \frac{1}{1 + \frac{1}{1 + \frac{1}{2}}} = \frac{1}{1 + \frac{1}{\frac{3}{2}}} = \frac{1}{1 + \frac{2}{3}} = \frac{1}{\frac{5}{3}} = \frac{3}{5} = \frac{3}{5} \times \frac{3}{1} = \frac{9}{5} \checkmark$$

$$B = \frac{2 + \frac{2+a}{2-a}}{2 - \frac{2+a}{2-a}} = \frac{\frac{2}{1} + \frac{2+a}{2-a}}{\frac{2}{1} - \frac{2+a}{2-a}} = \frac{\frac{2(2-a) + 2+a}{2-a}}{\frac{2(2-a) - 2-a}{2-a}} = \frac{\frac{4-2a+2+a}{2-a}}{\frac{4-2a-2-a}{2-a}} = \frac{6-a}{2-3a} \checkmark$$

Question 14:

$$A = \frac{\left(\frac{1}{a} + \frac{1}{b} - \frac{x}{ab}\right)(x+a+b)}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{2}{ab} - \frac{x^2}{a^2 b^2}} = \frac{\frac{x}{a} + 1 + \frac{b}{a} + \frac{x}{b} + \frac{a}{b} + 1 - \frac{x^2}{ab} - \frac{x}{b} - \frac{x}{a}}{\frac{b^2 + a^2 + 2ab - x^2}{a^2 b^2}} = \frac{2 + \frac{b}{a} + \frac{a}{b} - \frac{x^2}{ab}}{\frac{b^2 + a^2 + 2ab - x^2}{a^2 b^2}} = \frac{\frac{2ab + b^2 + a^2 - x^2}{ab}}{\frac{b^2 + a^2 + 2ab - x^2}{a^2 b^2}} = \frac{2ab + b^2 + a^2 - x^2}{ab} \times \frac{a^2 b^2}{b^2 + a^2 + 2ab - x^2} = \frac{(a^2 b^2)(\cancel{2ab + b^2 + a^2 - x^2})}{(ab)(\cancel{2ab + b^2 + a^2 - x^2})} = \frac{a^2 b^2}{ab} = ab \checkmark$$

$$B = \frac{2x+3}{2(2x-3)} + \frac{12x}{9-4x^2} + \frac{3-2x}{4x+6} = \frac{2x+3}{2(2x-3)} + \frac{-12x}{(2x-3)(2x+3)} + \frac{3-2x}{2(2x+3)}$$

$$= \frac{(2x+3)(2x-3) - 24x + (3-2x)(2x-3)}{2(2x-3)(2x+3)}$$

$$= \frac{4x^2 - 6x + 6x - 9 - 24x + 6x - 9 - 4x^2 + 6x}{2(2x-3)(2x+3)}$$

$$= \frac{-24x - 18}{2(2x-3)(2x+3)}$$

$$= \frac{-2(12x+9)}{2(2x-3)(2x+3)}$$

$$= -\frac{12x+9}{(2x-3)(2x+3)} = -\frac{12x+9}{4x^2-9} = 0 \checkmark$$

Question 15:

7840	2
3920	2
1960	2
980	2
490	2
245	5
49	7
7	7
1	

Donc $7840 = 2^5 \times 5 \times 7^2$ ✓

Question 16:

$$A = \frac{-3\left(\frac{2}{3}\right)^2 + 8\left(\frac{7}{2}\right)^2}{5\left(\frac{2}{5}\right)^2 - 6\left(\frac{4}{3}\right)^2} = \frac{-3 \times \frac{4}{9} + 8 \times \frac{49}{4}}{5 \times \frac{4}{25} - 6 \times \frac{16}{9}} = \frac{-\frac{12}{9} + \frac{392}{4}}{\frac{20}{25} - \frac{96}{9}} = \frac{\frac{290}{3}}{-\frac{148}{15}} = \frac{290}{3} \times \left(-\frac{15}{148}\right) = -\frac{725}{24}$$
 ✓

Question 17:

$$12^3 \times 3^3 = (12 \times 3)^3$$

$$= 36^3$$
 ✓

$$125^2 \times 3^6 = (5 \times 25)^2 \times 3^6$$

$$= 5^2 \times 25^2 \times 3^6$$

$$= 5^2 \times (5^2)^2 \times 3^6$$

$$= 5^2 \times 5^4 \times 3^6$$

$$= 5^6 \times 3^6$$

$$= (5 \times 3)^6$$

$$= 15^6$$
 ✓

$$3^3 \times 5^6 = 3^3 \times (5^2)^3$$

$$= 3^3 \times 25^3$$

$$= (3 \times 25)^3$$

$$= 75^3$$
 ✓

$$7^2 \times 2^3 = \text{Impossible}$$
 ✓

Question 18:

Entraînement au calcul algébrique : corrigé.

Correction de la question 1. 1°) $\frac{1}{(-1)^n} = (-1)^n$

2°) $(-1)^{n+2} = (-1)^n(-1)^2 = (-1)^n$

3°) $(-1)^{2n} = ((-1)^2)^n = 1^n = 1$

4°) $(-1)^{2n+1} = (-1)(-1)^{2n} = -1$ grâce à la question précédente

Correction de la question 2. 1°)

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

$$(a+b)(a-b) = a^2 - b^2$$

2°)

$$\begin{aligned} (a+b+c)^2 &= (a+b+c)(a+b+c) \\ &= a^2 + ab + ac + ba + b^2 + bc + ca + cb + c^2 \\ &= \boxed{a^2 + b^2 + c^2 + 2(ab + ac + bc)} \end{aligned}$$

Correction de la question 3. 1°) $(a-b)(a^2+ab+b^2) = a^3 + a^2b + ab^2 - ba^2 - ab^2 - b^3 = \boxed{a^3 - b^3}$.

2°) $a^3 + b^3 = a^3 - (-b^3) = a^3 - (-b)^3 = (a - (-b))(a^2 + a(-b) + (-b)^2) = \boxed{(a+b)(a^2 - ab + b^2)}$.

3°) $27x^3 + 8 = 3^3x^3 + 2^3 = (3x)^3 + 2^3 = (3x+2)((3x)^2 - 3x \times 2 + 2^2) = \boxed{(3x+2)(9x^2 - 6x + 4)}$. Le discriminant du trinôme qui apparaît est strictement négatif, on ne factorise pas plus dans $\mathbb{R}$.

Correction de la question 4. 1°) $A = x^4 - x^2 = x^2x^2 - x^2 = x^2(x^2 - 1) = \boxed{x^2(x-1)(x+1)}$

2°)

$$\begin{aligned} B &= x^2 - 2x + 1 - (x-1)(2x+3) \\ &= (x-1)^2 - (x-1)(2x+3) \\ &= (x-1)(x-1-2x-3) \\ &= (x-1)(-x-4) = \boxed{(1-x)(x+4)} \end{aligned}$$

3°)

$$\begin{aligned} C &= (3x+2+x-1)(3x+2-(x-1)) \\ &= \boxed{(4x+1)(2x+3)} \end{aligned}$$

4°)

$$\begin{aligned} D &= x^2(x+1) + x + 1 \\ &= \boxed{(x^2+1)(x+1)} \end{aligned}$$

5°)

$$\begin{aligned} E &= (3x)^2 - 7^2 + (3x+7)(2x+3) \\ &= (3x+7)(3x-7) + (3x+7)(2x+3) \\ &= (3x+7)(3x-7+2x+3) \\ &= \boxed{(3x+7)(5x-4)} \end{aligned}$$

Correction de la question 5. 1°) $\frac{7}{8} + \frac{5}{12} - \frac{4}{3} = \frac{7 \times 3 + 5 \times 2 - 4 \times 8}{24} = -\frac{1}{24}$

2°) $\frac{1}{(n+1)^2} + \frac{1}{n+1} - \frac{1}{n} = \frac{n + n(n+1) - (n+1)^2}{n(n+1)^2} = -\frac{1}{n(n+1)^2}$

Correction de la question 6.

$$x = \frac{\frac{a}{\frac{b}{\frac{c}{d}}}}{\frac{c}{d}} = \frac{a}{b} \frac{1}{\frac{c}{d}} = \frac{a}{b} \frac{d}{c} = \boxed{\frac{ad}{bc}}.$$

$$y = \frac{\frac{a}{\frac{b}{\frac{c}{d}}}}{\frac{c}{d}} = \frac{a}{bd} = \boxed{\frac{ac}{bd}}.$$

$$z = \frac{\frac{a}{\frac{b}{\frac{c}{d}}}}{\frac{c}{d}} = \frac{ac}{bd} = \boxed{\frac{ac}{bd}}.$$

$$t = \frac{\frac{a}{\frac{b}{\frac{c}{d}}}}{\frac{c}{d}} = \frac{a}{bcd} = \boxed{\frac{a}{bcd}}.$$

$$A = \frac{\frac{a^2}{\frac{3b}{ac}}}{\frac{6b}{6b}} = \frac{6ba^2}{3bac} = \boxed{\frac{2a}{c}}.$$

$$B = \frac{\frac{\frac{3}{10} \times \frac{15}{9}}{\frac{2}{9} \times \frac{5}{15}}}{\frac{5 \times 2}{3 \times 3} \times \frac{5 \times 3}{3 \times 3} \times 2} = \frac{1}{\frac{3}{2}} = \boxed{\frac{2}{3}}.$$

$$C = \frac{6(3-\frac{1}{2})(4+\frac{1}{3})}{12(5+\frac{1}{4})(7-\frac{1}{3})} = \frac{\frac{5}{2} \frac{13}{3}}{\frac{21}{4} \frac{20}{3}} = \frac{\frac{5 \times 13}{2 \times 3}}{\frac{21 \times 4 \times 5}{2 \times 3}} = \frac{13}{21 \times 4} = \boxed{\frac{13}{84}}.$$

Correction de la question 7. $A = \frac{x+1}{(x-1)(x+1)} - \frac{x-1}{(x+1)(x-1)} + \frac{2x}{1-x^2}$
 $= \frac{x+1-x+1}{x^2-1} - \frac{2x}{x^2-1} = \frac{2-2x}{x^2-1} = \frac{-2(x-1)}{(x-1)(x+1)} = \frac{-2}{x+1}.$

$B = \frac{b(x-a)}{abx} + \frac{x(a-b)}{abx} + \frac{a(b-x)}{axb} = \frac{bx-ba+xa-xb+ab-ax}{abx} = 0$

Correction de la question 8. $\frac{14^2 \times 9^2}{3^5 \times 7} = \frac{7^2 \times 2^2 \times (3^2)^2}{3^5 \times 7} = \frac{7 \times 4 \times 3^4}{3^5} = \frac{7 \times 4}{3} = \boxed{\frac{28}{3}}$

Correction de la question 9. $x = \frac{2^3 3^2}{3^{-2} 2^{-2} 3^4 2^8} = \frac{2^3 3^2}{3^2 2^6} = \boxed{2^{-3}}$

$y = 2^{100} + 2^{101} = 2^{100}(1+2) = \boxed{3 \cdot 2^{100}}$

$z = 2^{101} - 2^{100} = 2^{100}(2-1) = \boxed{2^{100}}$

$t = \boxed{2 \cdot 3^{15}}$

$u = \frac{(3^2(-2)^4)^8}{((-3)^5 2^3)^{-2}} = 3^{16} 2^{32} ((-3)^5 2^3)^2 = 3^{16} 2^{32} 3^{10} 2^6 = \boxed{2^{38} 3^{26}}$

Correction de la question 10.

$A = 3x^2 y^3 - y(xy)^2 = 3x^2 y^3 - yx^2 y^2 = 3x^2 y^3 - x^2 y^3 = \boxed{2x^2 y^3}.$

$B = \frac{4x^2 y^3 - (xy)^2 y}{x^2 y^2 \times (-x)^3} = \frac{4x^2 y^3 - x^2 y^3}{-x^2 y^2 x^3} = -\frac{3x^2 y^3}{x^5 y^2} = -\frac{3x^2 y^2 \times y}{x^2 y^2 \times x^3} = \boxed{-\frac{3y}{x^3}}.$

$C = \frac{(-a)^7 \times (-b^2 c^3)^3}{-b^3 c \times (-a)^5} = \frac{(-a^7) \times (-b^6 c^9)}{-b^3 c \times (-a^5)} = \frac{a^7 b^6 c^9}{a^5 b^3 c} = \boxed{a^2 b^3 c^8}.$

Correction de la question 11. $A = \frac{4}{3-\sqrt{5}} = \frac{4(3+\sqrt{5})}{9-5} = \boxed{3+\sqrt{5}}$

$B = \frac{1}{\sqrt{\sqrt{2}}} \sqrt{\frac{1+\sqrt{2}}{2}} = \sqrt{\frac{1+\sqrt{2}}{2\sqrt{2}}} = \sqrt{\frac{\sqrt{2}+2}{4}} = \boxed{\frac{\sqrt{\sqrt{2}+2}}{2}}$

$C = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} - \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} = \frac{(\sqrt{3}+\sqrt{2})^2}{3-2} - \frac{(\sqrt{3}-\sqrt{2})^2}{3-2}$
 $= 3+2+2\sqrt{6} - (3+2-2\sqrt{6}) = \boxed{4\sqrt{6}}$

Correction de la question 12.

$A = \frac{\frac{1}{1+x} + \frac{1-x}{(1+x)^2}}{\sqrt{\frac{1-x}{1+x}} \left(1 + \frac{1-x}{1+x}\right)} = \frac{\frac{1+x+1-x}{(1+x)^2}}{\frac{\sqrt{1-x}}{\sqrt{1+x}} \frac{1+x+1-x}{1+x}} = \frac{2}{(1+x)^2} \frac{1}{\frac{2\sqrt{1-x}}{(1+x)^{\frac{3}{2}}}} = \frac{1}{(1+x)^{\frac{1}{2}}} \frac{1}{\sqrt{1-x}}$

$A = \frac{1}{\sqrt{(1+x)(1-x)}} = \boxed{\frac{1}{\sqrt{1-x^2}}}$

Correction de la question 13. Pour $A = \frac{\frac{1}{1 + \frac{1}{1 + \frac{1}{2}}}}{\frac{1}{1 + \frac{1}{1 - \frac{1}{2}}}}$:

$$1 + \frac{1}{1 + \frac{1}{2}} = 1 + \frac{1}{\frac{3}{2}} = 1 + \frac{2}{3} = \frac{5}{3} \quad \text{et} \quad 1 + \frac{1}{1 - \frac{1}{2}} = 1 + \frac{1}{\frac{1}{2}} = 1 + 2 = 3.$$

Donc $A = \frac{\frac{5}{3}}{3} = \boxed{\frac{5}{9}}$.

$$B = \frac{2 + \frac{2+a}{2-a}}{2 - \frac{2+a}{2-a}} = \frac{\frac{2(2-a) + 2+a}{2-a}}{\frac{2(2-a) - 2-a}{2-a}} = \frac{4-2a+2+a}{4-2a-2-a} = \boxed{\frac{6-a}{2-3a}}$$

Correction de la question 14.

$$\begin{aligned} A &= \frac{\frac{b+a-x}{ab}(x+a+b)}{\frac{b^2+a^2+2ab-x^2}{a^2b^2}} \\ &= \frac{(b+a-x)(x+a+b)}{b^2+a^2+2ab-x^2} ab \\ &= ab \end{aligned}$$

$$\begin{aligned} B &= \frac{(2x+3)^2}{2(2x-3)(2x+3)} - \frac{24x}{2(2x-3)(2x+3)} + \frac{(3-2x)(2x-3)}{2(2x+3)(2x-3)} \\ &= \frac{(2x+3)^2 - 24x - (2x-3)^2}{2(2x-3)(2x+3)} \\ &= \frac{(2x+3 - (2x-3))(2x+3 + 2x-3) - 24x}{2(2x-3)(2x+3)} \\ &= \frac{6.4x - 24x}{2(2x-3)(2x+3)} \\ &= 0 \end{aligned}$$

Correction de la question 15. $7840 = 10 \times 784 = 2 \times 5 \times 2 \times (350 + 42) = 2^2 \times 5 \times 2 \times (175 + 21) = 2^3 \times 5 \times 196 = 2^3 \times 5 \times 2 \times 98 = 2^4 \times 5 \times 2 \times 49 = \boxed{2^5 \times 5 \times 7^2}$

Correction de la question 16. $A = \frac{-3\left(\frac{2}{3}\right)^2 + 8\left(\frac{7}{2}\right)^2}{5\left(\frac{2}{5}\right)^2 - 6\left(\frac{4}{3}\right)^2} = \frac{-\frac{4}{3} + 2 \times 7^2}{\frac{4}{5} - 2\frac{16}{3}} = \frac{\frac{2}{3}(-2 + 3 \times 49)}{\frac{4}{15}(3 - 5 \times 8)} = \frac{\frac{2}{3} \times 145}{\frac{4}{15} \times (-37)} =$
 $-\frac{2 \times 145 \times 15}{4 \times 3 \times 37} = -\frac{145 \times 5}{2 \times 37} = \boxed{-\frac{725}{74}}$

Correction de la question 17.

$$12^3 \times 3^3 = (12 \times 3)^3 = \boxed{36^3}$$

$$125^2 \times 3^6 = (5^3)^2 \times 3^6 = 5^6 \times 3^6 = \boxed{15^6}$$

$$3^3 \times 5^6 = 3^3 \times (5^2)^3 = (3 \times 25)^3 = \boxed{75^3}.$$

$7^2 \times 2^3$ ne peut pas s'écrire sous la forme voulue.

Correction de la question 19. 1°)

$$\begin{aligned} A &= 16(2x+7)^2 - 25(3x-7)^2 = (4(2x+7))^2 - (5(3x-7))^2 \\ &= (4(2x+7) - 5(3x-7))(4(2x+7) + 5(3x-7)) \\ &= (-7x+63)(23x-7) = \boxed{7(9-x)(23x-7)} \end{aligned}$$

2°)

$$\begin{aligned} B &= 9x^2(2x+1) - (2x+1) \\ &= (9x^2-1)(2x+1) \\ &= ((3x)^2-1)(2x+1) \\ &= \boxed{(3x-1)(3x+1)(2x+1)} \end{aligned}$$

3°)

$$\begin{aligned} C &= (4x^2-25)(x+2) - (x^2-4)(2x+5) + (5x+10)(2x+5) \\ &= (2x-5)(2x+5)(x+2) - (x-2)(x+2)(2x+5) + 5(x+2)(2x+5) \\ &= (2x+5)(x+2)(2x-5-(x-2)+5) \\ &= (2x+5)(x+2)(x+2) = \boxed{(2x+5)(x+2)^2} \end{aligned}$$

Correction de la question 20. Comme 4 et 5 sont strictement positifs, $\frac{7}{4} < \frac{9}{5} \iff 7 \times 5 < 9 \times 4$ ie $35 < 36$, ce qui est bien le cas donc on a bien : $\frac{7}{4} < \frac{9}{5}$.

Correction de la question 21. Comme $1 + \sqrt{2}$ et $\sqrt{3}$ sont positifs, il suffit de comparer leurs carrés. $(1 + \sqrt{2})^2 = 1 + 2 + 2\sqrt{2} = 3 + 2\sqrt{2} > \sqrt{3}^2$. Donc, $1 + \sqrt{2} > \sqrt{3}$.

Correction de la question 22. Calculons $\left(\frac{1+\sqrt{5}}{4}\right)^2 = \frac{1+5+2\sqrt{5}}{16} = \frac{3+\sqrt{5}}{8}$.

Comme $\frac{1+\sqrt{5}}{4}$ est un nombre positif, on a bien $\sqrt{\frac{3+\sqrt{5}}{8}} = \frac{1+\sqrt{5}}{4}$.

Correction de la question 23. Justifions d'abord que A existe.

$(4\sqrt{3})^2 = 48 < 7^2 = 49$ donc $4\sqrt{3} < 7$ donc $7 - 4\sqrt{3} > 0$.

Calculons A^2 pour commencer.

$$\begin{aligned} A^2 &= \left(\sqrt{7-4\sqrt{3}} - \sqrt{7+4\sqrt{3}} \right)^2 \\ &= 7 - 4\sqrt{3} + 7 + 4\sqrt{3} - 2\sqrt{7-4\sqrt{3}}\sqrt{7+4\sqrt{3}} \\ &= 14 - 2\sqrt{(7-4\sqrt{3})(7+4\sqrt{3})} = 14 - 2\sqrt{49-48} = \boxed{12} \end{aligned}$$

$0 \leq 7 - 4\sqrt{3} < 7 + 4\sqrt{3}$ donc $\sqrt{7-4\sqrt{3}} < \sqrt{7+4\sqrt{3}}$. Ainsi, $A < 0$.

On en déduit que : $\boxed{A = -\sqrt{12} = -2\sqrt{3}}$.

Correction de la question 24. $e^{3\ln 2} = 2^3 = 8$.

Correction de la question 25. Cette expression existe sur $\mathbb{R}^*$.

$\forall x \in \mathbb{R}^*, \ln(x^2) = \ln(|x|^2) = 2\ln(|x|)$ car $|x| > 0$.

Correction de la question 26.

$$\begin{aligned} 2\ln\left(\frac{3}{4}\right) - 3\ln\left(\frac{3}{8}\right) &= 2\ln(3) - 2\ln(4) - 3\ln(3) + 3\ln(8) \\ &= -\ln(3) - 2\ln(2^2) + 3\ln(2^3) \\ &= -\ln(3) - 2.2\ln(2) + 3.3\ln(2) \\ &= 5\ln(2) - \ln(3) \\ \ln\left(\frac{1}{\sqrt{2}}\right) &= -\ln(\sqrt{2}) \\ &= -\frac{1}{2}\ln(2) \end{aligned}$$

Correction de la question 27. L'expression existe ssi $x > -1$.

$$e^{x-\ln(x+1)} = e^x e^{-\ln(x+1)} = e^x \frac{1}{e^{\ln(x+1)}} = \frac{e^x}{x+1}.$$